The Growth Effect of Opening a Stock Exchange

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Current Motivating Example: Ethiopia



Reuters

Ethiopia launches stock exchange in fresh step to liberalise economy

By **Dawit Endeshaw**

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ADDIS ABABA, Jan 10 (Reuters) - Ethiopia launched a securities exchange on Friday, in what officials touted as a milestone in Prime Minister Abiy Ahmed's drive to open up the country's tightly controlled economy to private investment.

Just one company, Wegagen Bank, was initially listed on the Ethiopian Security Exchange (ESX), although ESX chief executive Tilahun Kassahun said he expected 90 companies to eventually join over the next 10 years.

However, progess has been slowed by armed conflict in several parts of the country, including a 2020-2022 civil war in the northern Tigray region that killed hundreds of thousands. Some investors have also raised questions about the government's true commitment to liberalisation.

https://www.reuters.com/world/africa/ethiopia-launches-stock-exchange-fresh-step-liberalise-economy-2025-01-10/

What are the real effects of opening a stock exchange?

- Important question for policy makers and investors that has surprisingly little empirical evidence

- A number of papers document positive relationship between the development of existing exchanges and economic growth (Atje and Jovanovic (1993), Beck and Levine (2004), Levine and Zervos (1998))

- Minier (2009) innovates by focusing on the transformative event of *opening an exchange*
  - compares growth in opening countries vs non-opening countries
  - finds that opening countries modestly outgrow controls over five years, with ambiguous longer-term effects

- Bekaert et al. (2005) show that equity market liberalizations increase annual economic growth, with largest growth response occurring in countries with high-quality institutions

Research Questions

We assemble the most comprehensive data to-date on countries' first stock exchanges to ask:

- What is the effect of opening an exchange on economic growth?
- Does this vary as a function of the quality of the nation's
  - institutions (Glaeser et al. (2004), Acemoglu et al. (2001))
  - or initial income ("Convergence hypothesis", e.g., Barro and Sala-i-martin (1992), Pritchett (1997))
- Where does this effect occur within the country?

Preview of results

At a *national* level

- Opening a stock exchange leads to higher growth in GDP per capita (average 15-year effect of 17–18 pp)
- The effect is larger for countries with:
  - High governance (15-year effect of 22–23 pp vs. 11–13 pp for low governance)
  - Low pre-exchange income (15-year effect of 31–34 pp vs. −5 to 2.5 pp for high income)

At a *subnational* level

- 10-year "nightlight" growth increases by 28 pp (15-year: 40 pp) in the opening city relative to similar cities
- The effect is concentrated within 50 miles of the exchange

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Night lights: DMSP annual composites, 1992–2013, intercalibrated following Elvidge et al. (2009).

Real Per Capita GDP data from the Global Macro Database.

Other national economic variables come from WorldBank:

Literacy rate	Central government debt (% of GDP)
Exports (% of GDP)	Inflation rate
Interest rate	Agriculture share of GDP
Life expectancy	Gross domestic savings (% of GDP)
Population density	

Governance Matters Data

Kaufmann, Kraay, Mastruzzi (2005) construct perception-based indicators of governance across six dimensions:

Voice and Accountability	Political Stability
Government Effectiveness	Rule of Law
Control of Corruption	Regulatory Quality

209 countries; 37 data sources; 31 organizations.

We classify a country as *high governance* if the average of the six indicators in 1996 is above the sample median.

Exchange Openings

List of Exchange Openings is hand-compiled, and to our knowledge comprehensive: first exchanges in 152 countries, from Amsterdam (1602) to Ethiopia (2025).

Staggered DiD literature → units treatment history may affect their suitability as a control

We have to understand the history of exchanges in *each* country, not just those whose openings we study.

Pre-Modern Exchange Openings

Country Name	Year	Country Name	Year
Netherlands	1602	New Zealand	1866
France	1724	Uruguay	1867
Portugal	1769	Belgium	1872
Austria	1771	India	1875
United States	1790	Greece	1876
Ireland	1793	Japan	1878
United Kingdom	1801	Egypt, Arab Rep.	1883
Denmark	1808	South Africa	1887
Italy	1808	Brazil	1890
Germany	1815	Hong Kong SAR, China	1891
Norway	1819	Chile	1893
Spain	1831	Mexico	1894
Switzerland	1850	Sri Lanka	1894
Argentina	1854	Finland	1912
Peru	1860	Philippines	1927
Australia	1861	Luxembourg	1928
Canada	1861	Colombia	1928
Sweden	1863	Morocco	1929
Turkiye	1866	Pakistan	1947
		Venezuela, RB	1947

Modern Stock Exchange Openings (1953–1995)

Country	Year	Country	Year	Country	Year
Israel	1953	Kenya	1954	Korea, Rep.	1956
Bangladesh	1956	Nigeria	1961	Taiwan	1962
Iran	1968	Tunisia	1969	Jamaica	1969
Bermuda	1971	Kuwait	1972	Thailand	1975
Cote d'Ivoire	1976	Costa Rica	1976	Jordan	1978
Trinidad & Tobago	1981	Saudi Arabia	1985	Iceland	1986
Guatemala	1987	Barbados	1987	Oman	1989
Bolivia	1989	Bahrain	1989	Botswana	1989
Mauritius	1989	Russian Fed.	1990	Honduras	1990
Eswatini	1990	Ghana	1990	Panama	1990
Uzbekistan	1991	Dominican Rep.	1991	Croatia	1991
Iraq	1992	Malta	1992	Ukraine	1992
Slovak Rep.	1993	Bhutan	1993	Lithuania	1993
Montenegro	1993	Nepal	1994	Nicaragua	1994
Suriname	1994	Paraguay	1994	Zambia	1994
Kyrgyz Rep.	1995	Mongolia	1995		

Modern Stock Exchange Openings (1995–2018)

Country	Year	Country	Year	Country	Year
Sudan	1995	Moldova	1995	Albania	1996
Cyprus	1996	Kazakhstan	1996	Malawi	1996
North Macedonia	1996	Fiji	1996	Palestine	1997
Qatar	1997	Cayman Islands	1997	Uganda	1997
Algeria	1997	West Bank & Gaza	1997	Bulgaria	1997
Belarus	1998	Cabo Verde	1998	Tanzania	1998
Channel Islands	1998	Burkina Faso	1998	PN Guinea	1999
Mozambique	1999	Vietnam	2000	Faroe Islands	2000
UAE	2000	Georgia	2000	Azerbaijan	2000
Bahamas	2000	Cameroon	2001	Armenia	2001
Bosnia & Herz.	2002	Guyana	2003	Libya	2007
Haiti	2007	Maldives	2008	Curacao	2009
Syria	2009	Lao PDR	2011	Rwanda	2011
Cambodia	2012	Seychelles	2013	Tajikistan	2015
Somalia	2015	Myanmar	2015	Gibraltar	2015
Lesotho	2016	Angola	2018		

Stock Exchange Re-Openings

Country	Opened	Closed	Re-Opened
China	1920	1949	1990
Czechia	1871	1948	1993
Ecuador	1873	1900	1970
El Salvador	1965	1976	1992
Estonia	1920	1941	1996
Ethiopia	1965	1974	2025
Hungary	1864	1948	1990
Indonesia	1912	1942	1952
Latvia	1937	1940	1995
Lebanon	1920	1983	1996
Malaysia	1930	1946	1960
Namibia	1904	1910	1992
Poland	1817	1939	1991
Romania	1882	1945	1995
Serbia	1895	1941	1989
Singapore	1930	1946	1960
Slovenia	1924	1942	1990
Zimbabwe	1896	1924	1946

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Synthetic Controls (Abadie and Gardeazabal (2003))

X_o^1 : $(k \times 1)$ vector of pre-exchange characteristics for country o
X_j^0 : $(k \times J)$ matrix for J non-opening nations.

$$\min_{W_o} \sqrt{(X_o^1 - X_j^0 W_o)' V (X_o^1 - X_j^0 W_o)} \tag{1}$$

subject to $W_o \geq 0$ and $\sum_j W_{oj} = 1$.

Choose V^* to minimize:

$$(Y_o^1 - Y_j^0 W_o^*(V))' (Y_o^1 - Y_j^0 W_o^*(V)) \tag{2}$$

RMSPE Measures Fit Quality

$$\text{RMSPE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(Y_t - \hat{Y}_t\right)^2}$$

Synthetic Difference-in-Differences (SDID)

Introduced by Arkhangelsky et al. (2021)

Like synthetic controls of Abadie and Gardeazabal (2003), SDID picks a "portfolio" of control units to estimate counterfactual.

Rather than matching on the level and covariates, SDID matches along the trend of a single (outcome) variable. ▸ Details

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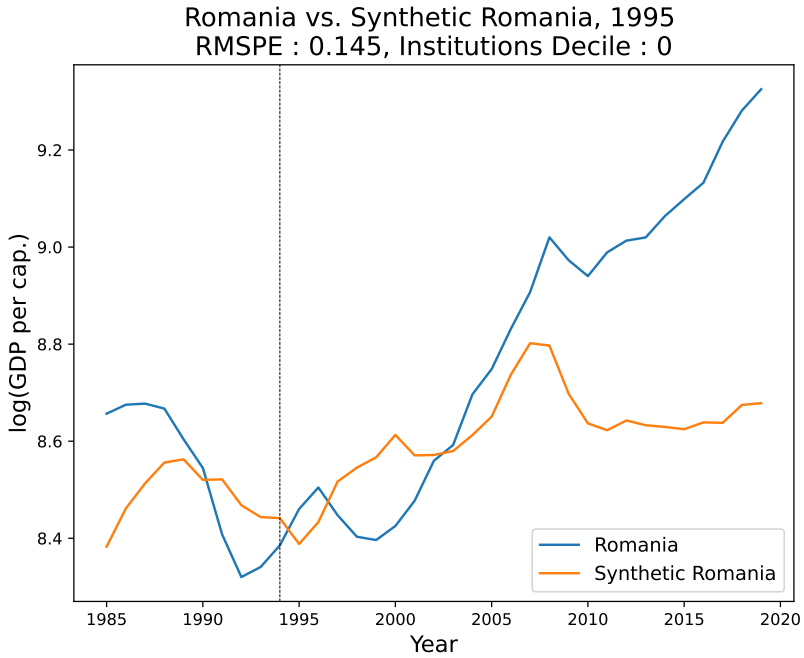
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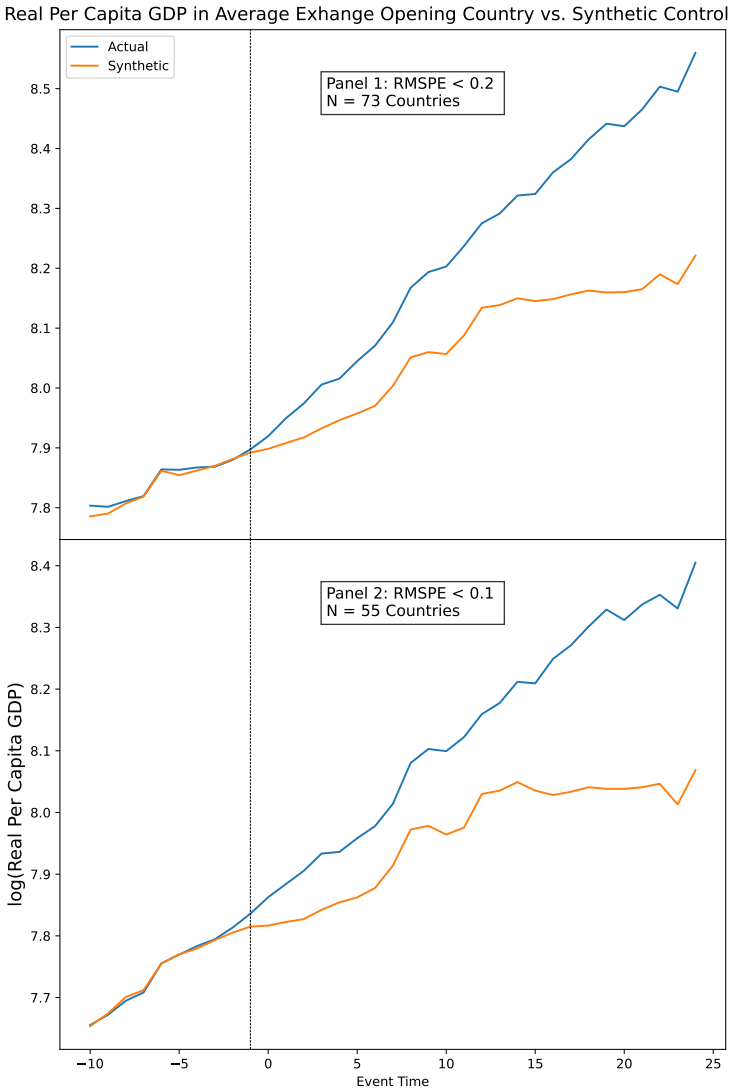
Romania - Synthetic Control Example



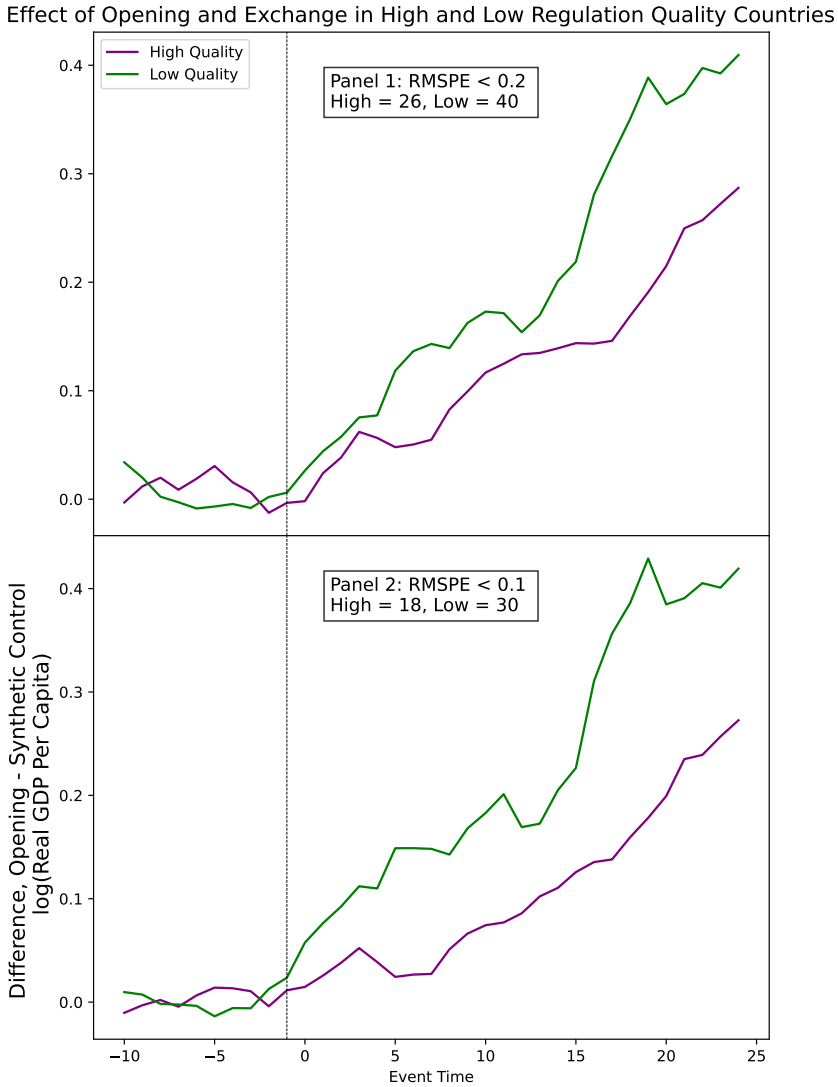
	count	mean	std	min	25%	50%	75%	max	Largest Weight
weights	42.0	0.02	0.11	0.0	0.0	0.0	0.0	0.69	ATG

	Romania	Synthetic Romania
gmsum	0.0	0.0
density	99.754	118.119
exports_GDP	23.569	93.643
lifeex	69.573	69.439
loggdppc	8.528	8.487
special.1.loggdppc	8.385	8.441

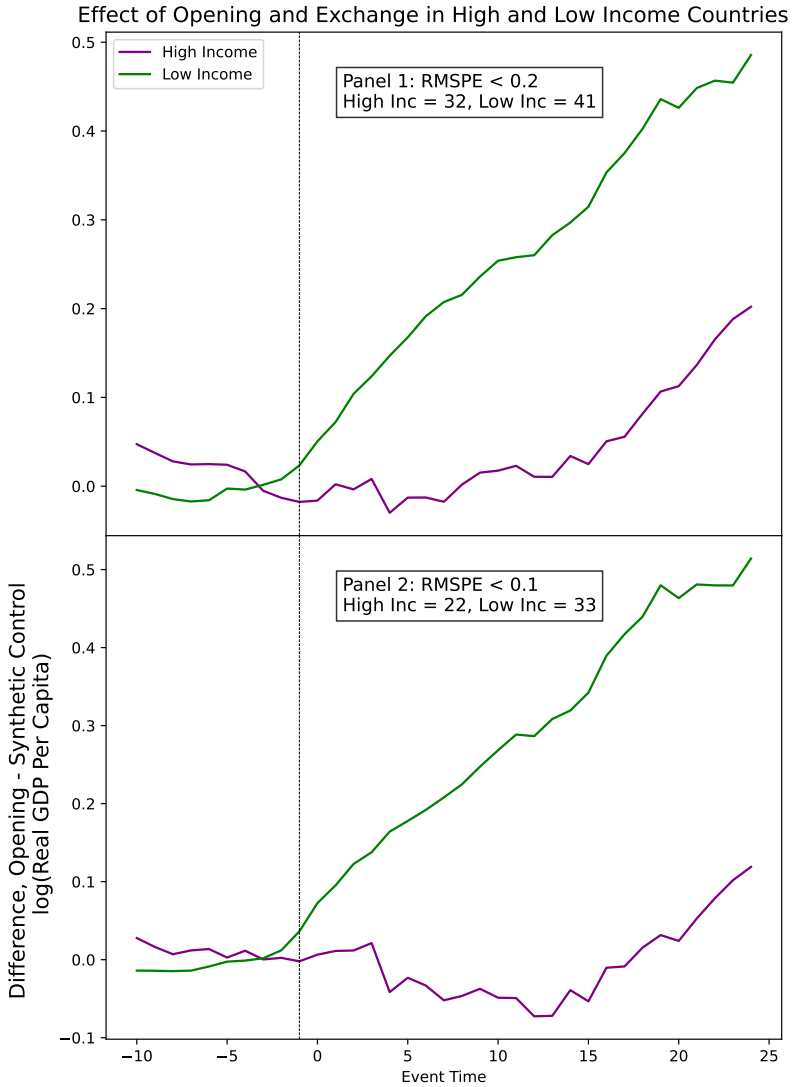
Average GDP vs Synthetic Controls



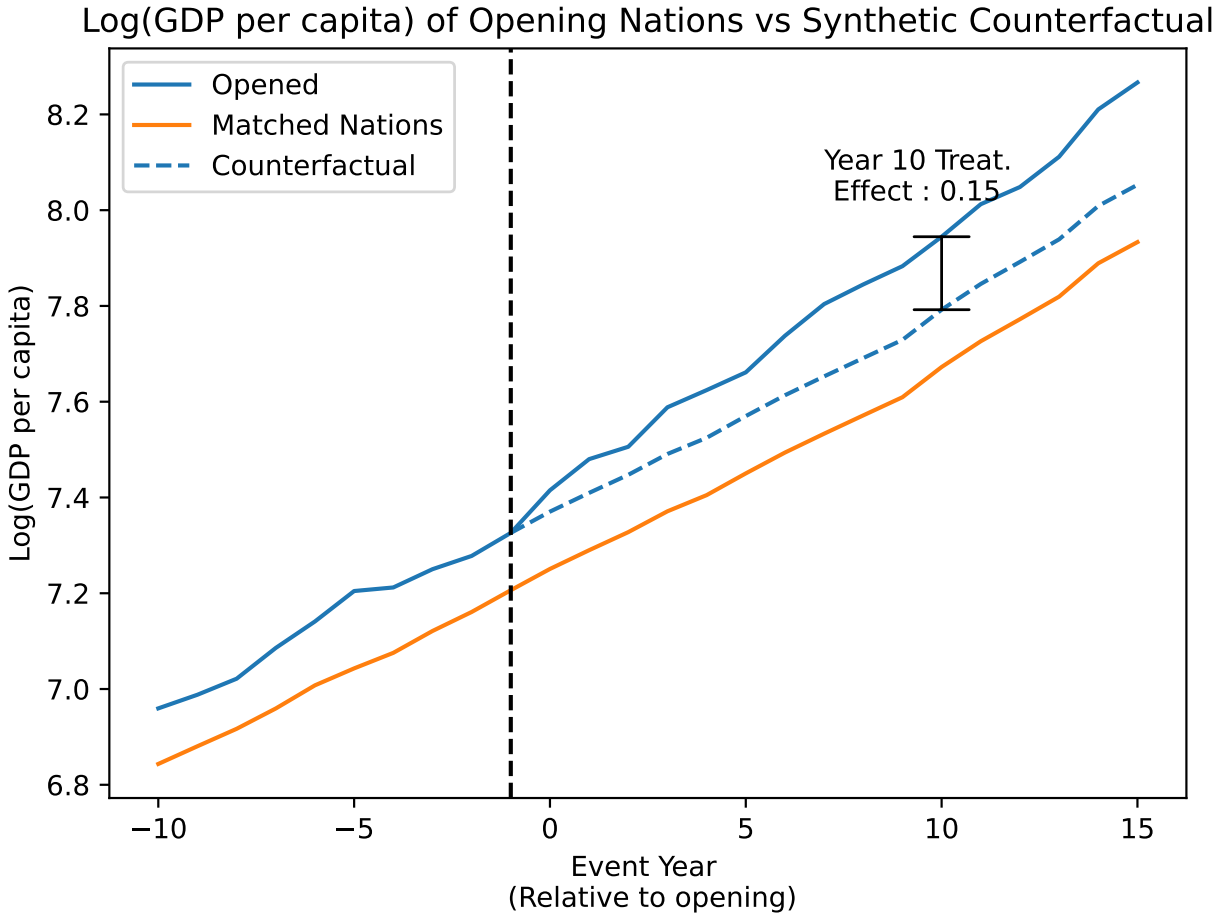
Average Effect by Governance (High vs. Low)



Average Effect by Income (High vs. Low)



SDID Approach Yields Similar Results



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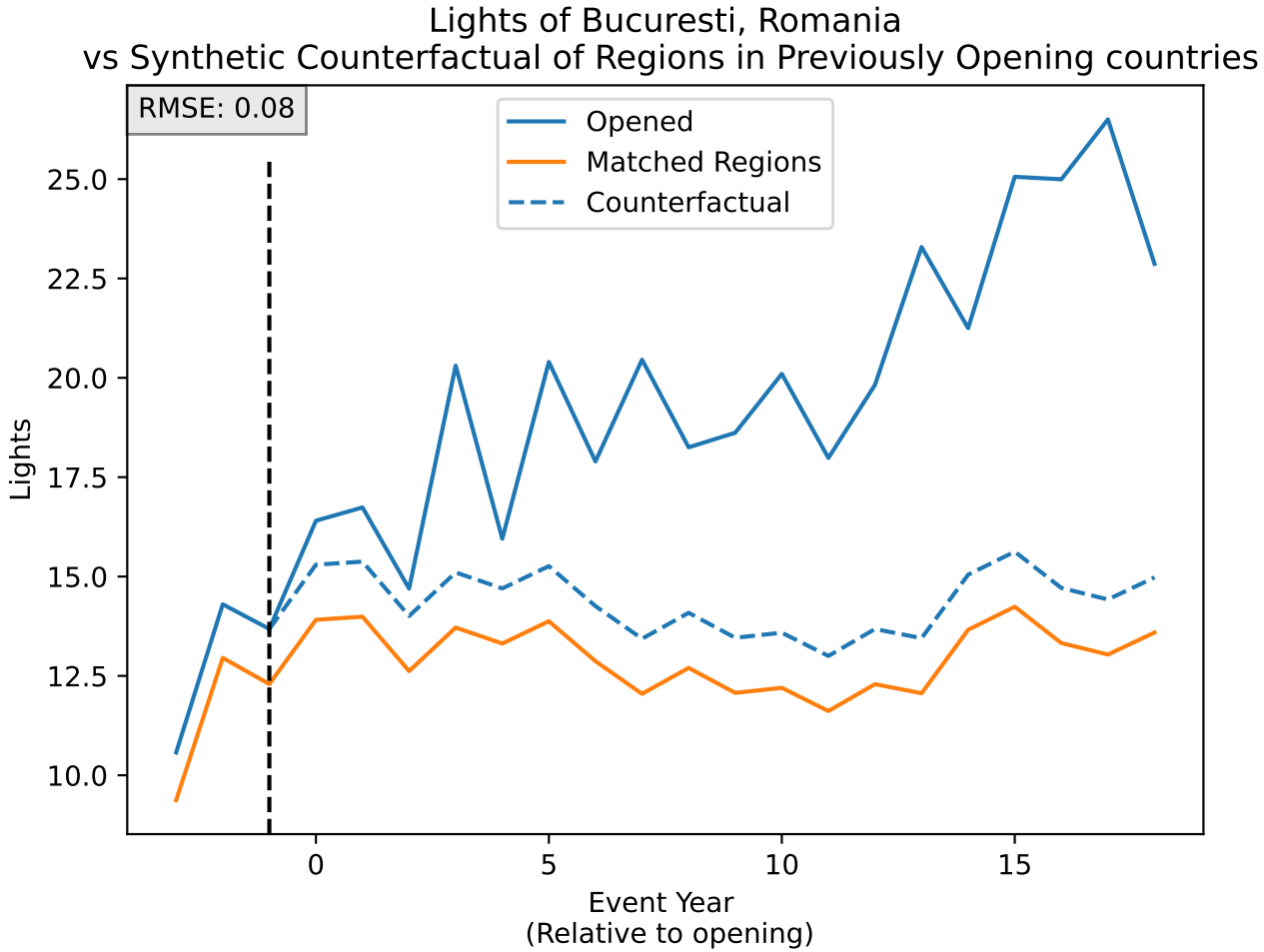
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Bucharest, Romania Example

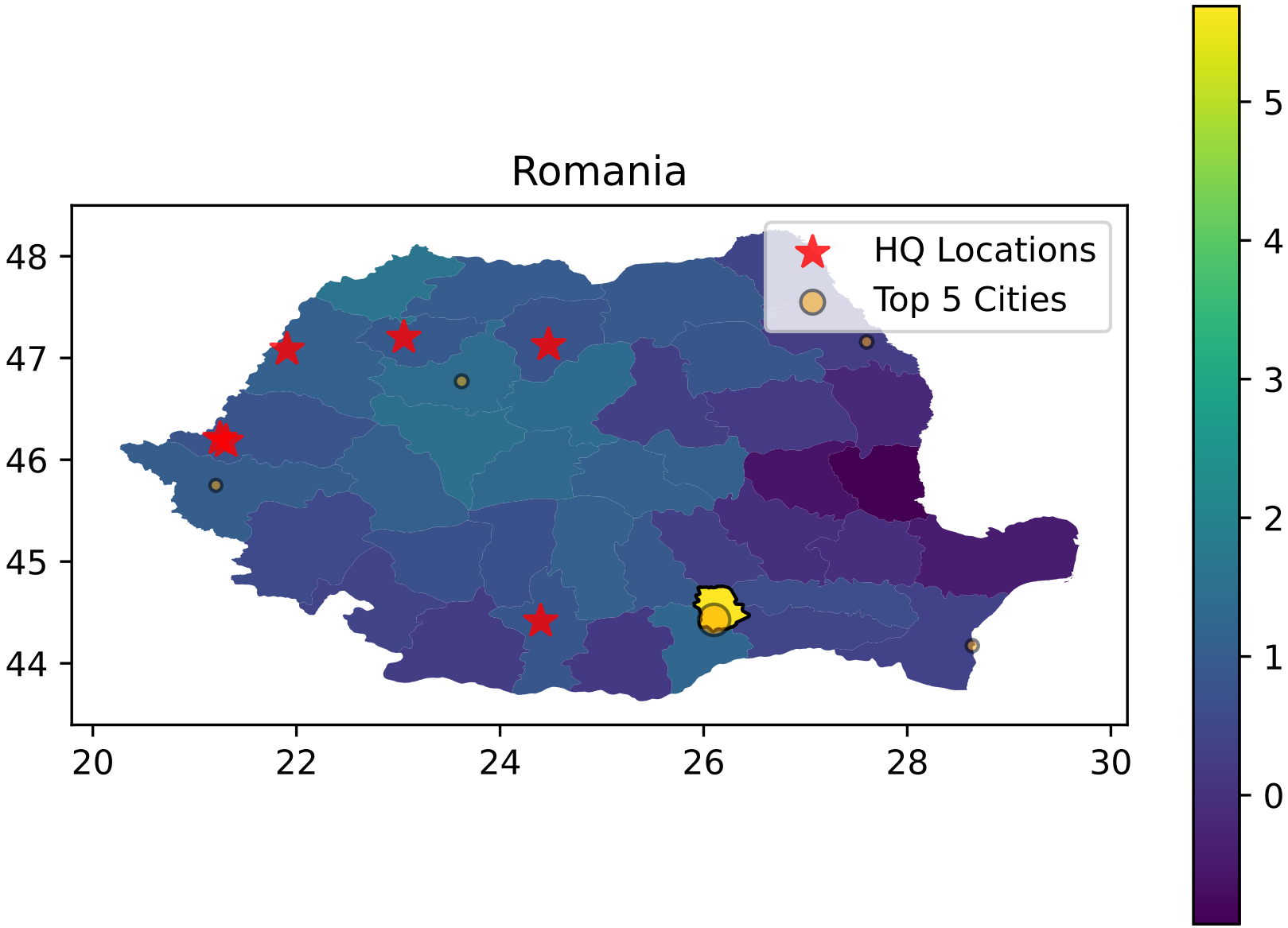


Effects of Opening an Exchange

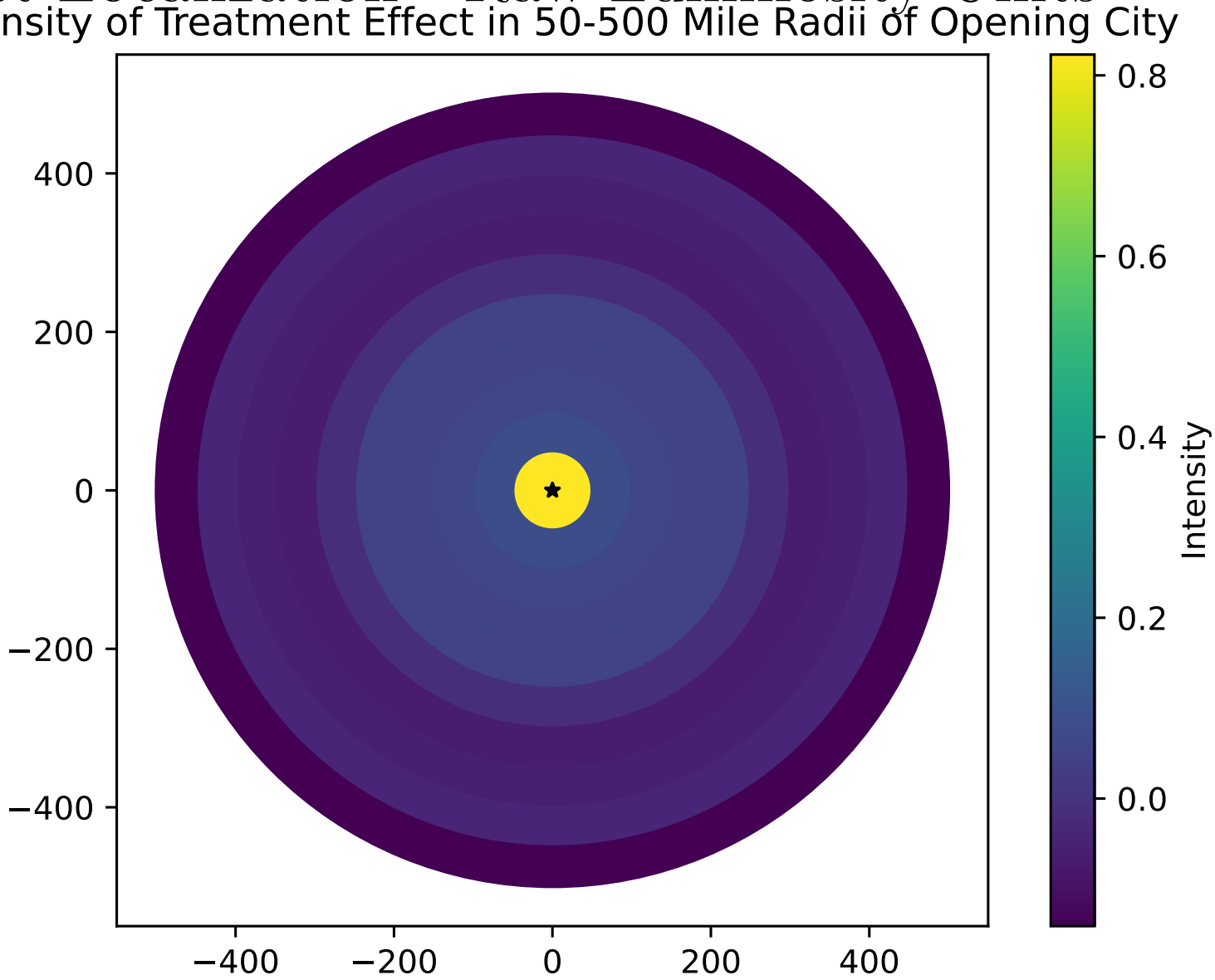
Country	Region	Year	Lights	Area	RMSE	0	3	5	7	10	15
Kyrgyzstan	Alamudun	1995	8.00	0.18	0.05	-3.2**	-2.6*	-5.0**	-1.7	-2.2	-2.3
Latvia	Rigas	1995	6.10	0.51	0.05	-0.7	0.2	-1.0	0.5	0.7	4.1*
Mongolia	Chingeltei	1995	13.30	0.01	0.02	-2.3*	1.1	2.1	4.3*	4.1	7.8**
Romania	Bucuresti	1995	16.40	0.16	0.08	1.1	5.2**	5.1**	7.0**	6.5*	9.4**
Sudan	Khartoum	1995	36.20	0.03	0.15	0.3	-2.6	0.4	4.4	3.9	10.3*
Albania	Tirane	1996	6.60	0.13	0.17	1.2	2.0*	4.5**	5.1**	7.1**	8.2**
Cyprus	Nicosia	1996	8.70	0.27	0.02	0.1	1.2	1.4	1.5	3.0	2.9
Estonia	Tallinna linn	1996	28.60	0.03	0.14	-0.2	7.3**	6.7*	7.8**	11.9**	6.5
Fiji	Rewa	1996	1.80	0.02	0.02	0.0	-0.1	-0.0	-0.1	-0.1	-0.8
Malawi	Blantyre	1996	3.10	0.17	0.01	0.1	0.4	0.6	0.5	1.0	0.2
Macedonia	Cair	1996	22.10	0.01	0.08	0.2	2.9	4.2	4.4	6.6	5.5
Bulgaria	Sofia	1997	10.70	0.15	0.03	-0.7	1.1	3.7*	2.2	2.4	7.3**
West Bank	Nablus	1997	19.50	0.06	0.05	1.4	1.3	1.1	1.6	2.0	3.8
Cape Verde	Santiago	1998	1.10	0.08	0.01	0.2	0.4	0.3	0.2	0.3	1.0
Tanzania	Dar-es-salaam	1998	9.80	0.13	0.04	-0.3	0.8	0.8	1.0	1.6	3.8
Mozambique	Maputo	1999	6.50	0.03	0.07	0.5	2.3**	2.5**	3.8**	5.0**	–
PN Guinea	National Cap.	1999	17.60	0.01	0.08	0.0	1.8	0.8	1.2	-2.3	–
Azerbaijan	Baku	2000	5.20	0.20	0.11	-0.4	0.8	1.5	1.2	1.0	–
UA Emirates	Abu Dhabi	2000	4.60	5.28	0.02	-0.2	0.9	1.6*	1.3	1.3	1.8
UA Emirates	Dubai	2000	15.70	0.36	0.01	0.2	3.6**	6.0**	6.4**	9.0**	11.3**
Haiti	Port-Au-Prince	2007	4.60	0.06	0.12	0.4	0.7	1.9	–	–	–
Mean (good fit)						-.11	1.37	1.87	2.63	3.14	4.75
(% of Lights)						-.8%	13%	18%	22%	28%	40%

Good-fit matches only (RMSE $<$ .2). SDID donor pool: regions in countries that opened an exchange before 1970. Effects in raw luminosity units.

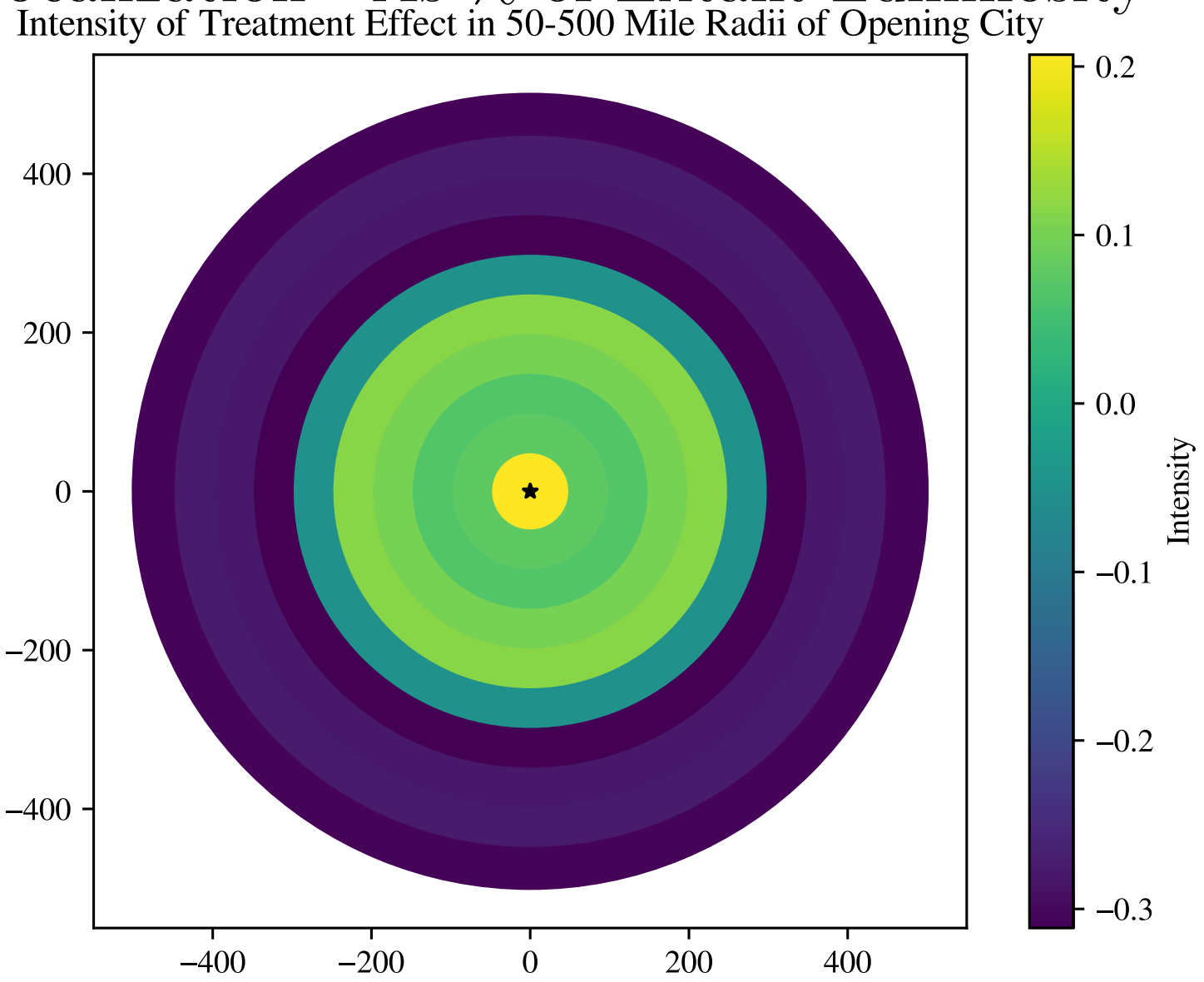
Romania Map Example



Effect Localization - Raw Luminosity Units



Effect Localization - As % of Extant Luminosity



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Conclusions

- **Do stock exchanges spur economic growth?** We find large and persistent increases in per capita GDP following exchange openings—17 to 18 percentage points over fifteen years on average.
- Effects are concentrated in countries with **strong institutions** and **low initial income**.
- **The growth is local.** Satellite data on nighttime lights show that growth is heavily concentrated within 50 miles of the exchange-opening city.
- **Causality?** Cannot argue that opening an exchange is exogenous. But, our synthetic controls approach and subnational data at least mitigate endogeneity concerns.

SDID - Regression

Standard TWFE Regression:

$$(\hat{\tau}^{\text{did}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\alpha,\beta,\mu,\tau} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} \left(Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau\right)^2 \right\} \quad (3)$$

A re-weighted version:

$$(\hat{\tau}^{\text{did}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\alpha,\beta,\mu,\tau} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} \left(Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau\right)^2 \hat{\omega}_i^{\text{sdid}} \hat{\lambda}_t^{\text{sdid}} \right\} \quad (4)$$

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SDID - Unit Weights

The unit weights $\hat{\omega}_i^{\text{sdid}}$ are given by:

$$(\hat{\omega}_0, \hat{\omega}^{\text{did}}) = \arg \min_{\omega_0 \in \mathbb{R},\, \omega \in \Omega} \left\{ \sum_{t=1}^{T_{\text{pre}}} \left(\omega_0 + \sum_{i=1}^{N_{\text{co}}} \omega_i Y_{it} - \frac{1}{N_{\text{tr}}} \sum_{i=N_{\text{co}}+1}^{N} Y_{it} \right)^2 + \xi^2 T_{\text{pre}} \|\omega\|_2^2 \right\} \tag{5}$$

$$\Omega = \left\{ \omega \in \mathbb{R}_+^N : \sum_{i=1}^{N_{\text{co}}} \omega_i = 1, \quad \omega_i = \frac{1}{N_{\text{tr}}} \quad \forall i = N_{\text{co}}+1, \ldots, N \right\}$$

And satisfy:

$$\sum_{i=1}^{N_{\text{co}}} \hat{\omega}_i^{\text{sdid}} Y_{it} \approx \frac{1}{N_{\text{tr}}} \sum_{i=N_{\text{co}}+1}^{N} Y_{it} \quad \text{for all } t = 1, \ldots, T_{\text{pre}}$$

SDID - Time Weights

The time period weights $\hat{\lambda}_t^{\text{sdid}}$ are given by:

$$(\hat{\lambda}_0, \hat{\lambda}^{\text{sdid}}) = \arg \min_{\lambda_0 \in \mathbb{R},\, \lambda \in \Lambda} \left\{ \sum_{i=1}^{N_{\text{co}}} \left(\lambda_0 + \sum_{t=1}^{T_{\text{pre}}} \lambda_t Y_{it} - \frac{1}{T_{\text{post}}} \sum_{t=T_{\text{pre}}+1}^{T} Y_{it} \right)^2 \right\} \quad (6)$$

$$\Lambda = \left\{ \lambda \in \mathbb{R}_+^T : \sum_{t=1}^{T_{\text{pre}}} \lambda_t = 1, \quad \lambda_t = \frac{1}{T_{\text{post}}} \quad \forall t = T_{\text{pre}} + 1, \ldots, T \right\}$$

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